

On the revision of the knot-invariant manuscript

A letter to Edison Vázquez

Dear Edison,

Thank you for sending the manuscript and the Supplementary Information, and for explaining so carefully how your interpretation has changed. I appreciate the work behind these revisions. You have followed the evidence even when it required changing the central claim. That willingness to reconsider an interpretation gives me confidence in the work, and I would like us to preserve it as we finish the paper.

I think there is a worthwhile paper here. Its strongest question is easy to state: when we call a knot unusual, what feature of the knot are we actually noticing? Your computations show that different scores lead us to different populations. The next step is to make the mathematical meaning of those populations clearer and to ensure that the statistical comparisons support the conclusions we draw from them.

I would like a substantive revision before I can approve the present version for submission. A few issues affect the justification of the results; others concern how we explain them. I have set out the reasons below, together with a suggested order of work. I have gone into some detail so that you can see exactly what needs attention and how to proceed.

These comments concern the manuscript, supplement, and relevant portions of the public code. The entire computation has not been independently reproduced. Where an issue calls for a new calculation, I have tried to distinguish that request from a correction that can already be made on mathematical grounds.

Begin with the mirror experiment. In Supplement §S3.1, the invariance of the stored θ polynomial under simultaneous inversion of its variables is used to treat the θ view as unchanged when knots are mirrored. This step needs to be reconsidered. Substituting inverse variables into a polynomial and taking the mirror image of the knot are different operations. A symmetry under the first operation does not establish the transformation under the second.

The source paper by Bar-Natan and van der Veen distinguishes these questions. Conjecture 20 concerns symmetries of the polynomial variables, while Conjecture 21 states

$$\theta(\overline{K}) = -\theta(K).$$

The word *conjecture* matters: we should retain its stated status and avoid using this formula as a proved general theorem. It does, however, make it especially important to justify the mirror action used in our experiment. The current argument establishes a variable symmetry, and the public script `stage18B_random_mirror_selection.py` then holds both Alexander and Theta fixed. This does not establish that the experiment has mirrored all five representations correctly. Reference [1] below gives the relevant source.

Please start with several small knots and compute θ for the knot and its actual mirror using an independent implementation. Check the polynomial coefficients, including their signs. If the archived coordinates include a normalization that removes a sign or chooses a

representative, describe it explicitly. Once the transformation for our encoding is established, apply it consistently to the mirror experiment and rerun the affected summaries. Until then, the reported recovery percentages and Jaccard overlaps should be regarded as unverified mirror-robustness results. This issue does not, on its own, show that the original catalogue calculations are wrong.

There is also a terminology correction here. An *unoriented knot* is a knot whose orientation has been forgotten. It need not be equivalent to its mirror. If we intend to identify mirror pairs, please say that explicitly. A score can be a knot invariant while distinguishing a knot from its mirror. The property we are testing is independence from the choice of representative of a mirror pair. This distinction should be corrected in Supplement §S3.2 and in the conclusion.

State precisely what the statistical tests establish. The move from independently randomized votes to a joint randomization is a sensible correction. It preserves the dependence among views and the observed consensus size. There is still a separate question: within the cells used for randomization, is it reasonable to reassign the selected labels without changing their relationship to the outcome under the null hypothesis? This is the exchangeability assumption. Preserving vote dependence does not establish it.

The amplitude-grid table is particularly informative. For the canonical selection of 220 knots, the reported mean width is 2.9727. The corresponding null means are

Bins per view	Null mean	Observed minus null mean
20	2.9499	0.0228
10	2.8287	0.1440
5	2.6120	0.3607
2	2.4160	0.5567

The contrast changes by about a factor of 24. This deserves a central place in the explanation. It does not prove that detailed amplitude control eliminates the association: fine cells also immobilize many selected observations, so the comparisons draw their information from different subsets. It does mean that the large two-bin contrast cannot be presented as a stable estimate of an effect remaining after detailed amplitude adjustment.

Please report the observed-minus-null difference together with the number of selected observations that can move at each resolution. Explain that a median cell size of ten is a practical criterion for obtaining usable permutations; it is not a theorem guaranteeing adequate adjustment. The strongest statement currently available is an association under the specified stratified comparisons. A broader claim about information beyond amplitude requires further justification.

The continuous-amplitude analysis needs a related repair. As described in Supplement §S5.2, a regression predicts the outcome from norms and structural variables, and the selected labels are then randomized among residuals within structural strata. Obtaining predictions out of fold is helpful: each observation is predicted by a regression trained on other observations. But cross-fitting does not make the residuals exchangeable across different norms.

For example, suppose that

$$Y = m(Z) + \varepsilon, \quad \mathbb{E}(\varepsilon \mid Z) = 0,$$

where Z contains the adjustment variables. Even if we knew the conditional mean m exactly, residuals could have much larger variance in one part of the norm range. If the selected knots are concentrated there, a permutation that mixes them with lower-variance observations gives the wrong reference distribution. Imperfect estimation of m can also leave a systematic prediction error in the selected tail. Cross-fitting addresses reuse of observations in fitting; these are additional difficulties.

Please treat the present residual means as useful descriptive checks while we establish a calibrated test. One possible approach estimates both $\mathbb{E}(Y \mid Z)$ and the selection probability $\mathbb{E}(H \mid Z)$, where H is the selection indicator, and studies the standardized covariance of their residuals. Such methods require assumptions about regression accuracy, variance, and the sampling scheme; reference [2] provides a starting point. We should choose the test with statistical advice and explain why its assumptions are plausible here.

Whichever method we use, the calibration experiments should include uneven selection probabilities, changing residual variances, and regression errors concentrated in the selected tail. The synthetic experiment already reported is useful evidence for its particular data-generating model. It cannot establish calibration for every form of dependence present in the knot atlas. Until this is settled, please reduce the prominence of the residual-randomization p -value in the abstract.

Bring the methods into agreement with the computation. There is a straightforward correction to the aggregation definition. The manuscript says that the third-largest per-view score is used. The public code first converts each view’s scores to empirical percentiles and then takes the third-largest percentile. That conversion is important because numerical scores from different views have different scales. Please put it in the formula and specify whether the reference distribution comes from validation or test data. Ranking the test observations for a fixed-size descriptive comparison is legitimate, but it should be distinguished from applying a validation-calibrated threshold to a new observation.

Please also give the full definition of the residual Mahalanobis and isolation-forest baselines. In the public implementation, a further PCA projection compresses the residuals to at most 64 coordinates, fitted from a default sample of 40,000 training observations. This can discard unusual residual directions, so it is part of the method being compared. State the projection dimension, fitting sample, eigenvalue floor, and relevant tuning choices. A small sensitivity check should establish whether this additional compression materially changes the comparison.

Some of the comparative language also needs tightening. A larger observed effect for one score does not establish that its effect differs significantly from another score’s effect. In particular, a corrected volume result for relative error and a result that does not survive correction for the conditional score do not constitute a test of the difference between the two methods. Please use “largest observed reduction” for the descriptive finding. If an ordering of methods is an important conclusion, specify a direct comparison that accounts for overlapping selections, shared controls where relevant, and the different matching coverage. Report balance in the individual representation norms as well as their mean and maximum.

The same care applies to the two width distributions. The present acknowledgement that no formal between-score test was performed is appropriate. Please keep the interpretation at that level throughout the abstract and discussion.

Separate exploration from a fresh test of the result. Your account of how the main claim evolved is scientifically valuable. We should describe it accurately. Evaluating a newly chosen statistic once does not make it an independent confirmation if the data have already helped us choose the question. The supplement makes this clear for the Rasmussen–signature analysis. Please check that the description of the width analysis is equally precise.

For a stronger confirmation, I would like us to reserve a genuinely untouched crossing block or a separate collection of knots, fix the analysis before examining its outcomes, and then apply it once. If that cannot be done within the present revision, we should state plainly that the evidence concerns exploration and interpolation within this atlas. A useful additional split would keep identical invariant tuples together: otherwise different knots with the same input representation may appear in both training and test sets. This would tell us how much of the apparent generalization extends to new algebraic inputs.

Let the elementary geometry lead the exposition. There is a very simple identity that would make the paper easier to understand. Let P be the fitted orthogonal projector, and write a nonzero standardized vector as $x = ru$, with $\|u\| = 1$. Then

$$D(x) = \|(I - P)x\|^2 = r^2\|(I - P)u\|^2,$$

and, whenever the denominator floor is inactive,

$$\frac{D(x)}{\|x\|^2} = \sin^2 \angle(x, \text{im } P).$$

Thus raw error measures squared distance from the retained subspace, while relative error measures squared angular deviation. A conditional percentile asks how unusual that deviation is among vectors of comparable length. These statements concern the fitted geometry in standardized coordinates; they do not assert a scaling symmetry of the original knot polynomials.

Please include the identity and its one-line proof near the beginning. It explains what the scores are doing before the reader encounters the statistical comparisons. At an exactly fixed positive radius, raw and relative errors have the same ordering. Their difference arises when observations at different radii are compared.

This also suggests a change in interpretation. Calling the relative score an “overcorrection” assumes that removing its association with norm is the desired objective. The score has a clear angular meaning. We can simply report its negative association with norm and ask which knots it selects. Likewise, near-zero rank correlation between an individual conditional score and norm is a calibration diagnostic; it does not establish complete independence from amplitude. Dependence among the views can still make the final consensus depend on amplitude.

Give the width threshold a topological meaning. I think the most useful immediate addition is the Turaev-genus consequence. Established results give

$$w_{\text{Kh}}(K) - 2 \leq g_T(K) \leq \text{dalt}(K),$$

where g_T is the Turaev genus and dalt is the dealternating number. The first inequality bounds the spread of Khovanov homology; the second follows because each crossing change alters the genus of the diagram's Turaev surface by at most one. The relevant statements and attributions are recorded in reference [3].

For the reader, please briefly explain the terms. The Turaev surface of a diagram is constructed from its all- A and all- B resolutions, and $g_T(K)$ is the minimum genus of these surfaces over diagrams of K . The dealternating number is the minimum number of crossing changes needed to turn a diagram into an alternating diagram. It should be distinguished from the alternation number, whose target is an alternating knot type.

Once the interpretation and grading conventions of the archived free coordinates are verified, their number W_{Kh} of occupied diagonals is at most the homological width. Missing intermediate diagonals do not spoil this inequality. Therefore

$$W_{\text{Kh}}(K) \geq 4 \implies g_T(K) \geq 2, \quad \text{dalt}(K) \geq 2.$$

In particular, such knots cannot be almost alternating.

The reported proportions correspond to 48 of the 220 canonical no-Khovanov selections, nine of the 31 held-out conditional selections, and five of the 31 held-out relative selections. Please check these counts against the individual records and list the knots. This gives the reader a concrete consequence of the selected width threshold. We should attribute the underlying inequalities and explain that our contribution is their application to the selected population.

There is one important limit to the conclusion: this certifies a genus lower bound for those knots. It does not itself prove enrichment for the actual Turaev genus, because other knots below the width threshold may also have large Turaev genus. Keeping this distinction explicit will make the addition both useful and reliable.

Find out what the several views contribute. Both Alexander and Jones are specializations of HOMFLY-PT. With consistent normalizations, the four-view tuple (Δ, V, P, θ) therefore contains the same algebraic information as (P, θ) . Its several coordinate systems may still provide different and useful geometries. That is an interesting feature of the experiment, and it would be good to examine it directly.

Please compare the present procedure with HOMFLY-PT alone, θ alone, and the pair, using the same held-out population and final selection size. The question is whether the derived views help us use the available information more effectively. This is a modest additional experiment with a clear purpose.

There is also an exact observation. Once the numerical models are fixed, knots with the same (P, θ) tuple receive the same deterministic scores from these four views. Identifier-based tie-breaking can separate them at a selection boundary, but that separation has no mathematical content. Please inspect these fibers, meaning groups of knots with the same tuple, and check how ties are treated. Look for informative pairs with the same HOMFLY-PT polynomial but different θ or different Khovanov structure. An example should be included only if it actually occurs in the data and has been checked. A good pair would give the reader something mathematical to examine closely.

Build mirror invariance into a score. After the mirror representations have been corrected, there is a simple construction worth testing. For any fixed fitted score S , set

$$S_{\text{sym}}(K) = \frac{S(K) + S(\bar{K})}{2}.$$

Then $S_{\text{sym}}(\bar{K}) = S_{\text{sym}}(K)$ because mirroring twice returns the original knot. This proves the desired symmetry of the score. It does not depend on a percentage of membership recovery.

Please keep mirror pairs together in training and test splits and make the subsequent calibration respect the same symmetry. Average the scores with care: averaging the feature vectors themselves can erase information. For example, any feature that changes sign under mirroring would vanish under that feature average. Compare the resulting selections with the corrected original procedure and report what is retained or changed. This would give the paper a precise constructive answer to one of its own limitations.

Consider one further homological quantity. There is a natural way to measure information lost when passing from homology to its Euler characteristic. Write

$$h_{ij} = \dim_{\mathbb{Q}} \text{Kh}^{i,j}(K; \mathbb{Q}), \quad c_j = \sum_i (-1)^i h_{ij}.$$

The polynomial $\sum_j c_j q^j$ is the graded Euler characteristic. Its normalization must be matched to the stored Jones data. Define

$$E(K) = \frac{1}{2} \left(\sum_{i,j} h_{ij} - \sum_j |c_j| \right).$$

If a_j and b_j are the total ranks in even and odd homological degrees at quantum degree j , respectively, then

$$E(K) = \sum_j \min(a_j, b_j) \geq 0.$$

The proof is the elementary identity $a + b - |a - b| = 2 \min(a, b)$. Thus E counts cancellation in forming the Euler characteristic. It uses the actual homology ranks, not the standardized coordinates.

After the main corrections, please calculate this quantity as an exploratory endpoint for selection without Khovanov. Does the method identify knots whose homology contains more cancellation? This question is more informative than a generic claim of larger “structure.” We should check prior use of the quantity before making a novelty claim, and any observation from this additional exploration should be labelled accordingly. If it adds no useful distinction, a short record of the calculation will be enough; it need not enlarge the main paper.

Make the prose as direct as the mathematics. The English is generally grammatical. What makes it difficult is the number of abstract nouns and repeated explanations of what the paper does or does not claim. I would like the reader to meet the knots and the geometry earlier. A possible title is *Distance, angle, and anomaly in knot-invariant spaces*, if the revised paper takes that form.

Please begin with the question, explain the three geometric meanings of the scores, and then show the selected populations. Include at least one checked knot example with enough of its polynomial or homological data for the reader to see the difference. The present figures

mostly summarize the behavior of the methods. An actual example would help explain why the distinction matters mathematically. Enlarge the text in Figure 1 and the compressed tables, especially Supplement Table S13.

The following are examples of the language changes I have in mind. They should guide a continuous reading of the whole paper, rather than a mechanical substitution of words.

Current wording	Suggested treatment
"scientific phenotype"	Name the property: alternation, width, or volume.
"operationalizes a low-amplitude-weighted notion of anomaly"	"favors vectors of smaller norm"
"paper-facing randomization"	"randomization used in Table 4"
"frozen full-fit canonical regime"	"selection fitted on the complete atlas," after defining it
"localizes larger stored Khovanov structure"	"selects knots supported on more Khovanov diagonals"
"scale-contextual definition"	"comparison among vectors of similar norm"
"non-familywise-significant geometric shift"	"volume difference that does not survive multiple-testing correction"
"continuous-amplitude sensitivity"	"analysis adjusting for the continuous norms"
"representation hardness"	"reconstruction error," unless a broader meaning is defined
"mobility-qualified"	Give the number of observations that can move.
"relative error overcorrects"	Report its negative association with norm.
"nearly disjoint populations"	Specify the pairs; raw and Mahalanobis have Jaccard overlap about 0.51.
"high-precision core"	Describe the observed overlap; there is no true anomaly label.
"rather than to a nonexistent Supplementary table"	Delete this account of editorial history.

Please also shorten the numerical inventory in the abstract. Keep the dataset size and one central quantitative comparison, with its qualifications. The sequence of residual means, tail proportions, sample sizes, and p -values currently makes it difficult to retain the idea. State each limitation where it first becomes relevant and avoid repeating it throughout the introduction, results, discussion, and conclusion. Words such as "scale-neutral," "stronger," and "overcorrects" should either receive a precise meaning or be replaced by the actual observation.

How I would organize the remaining work. To keep the revision manageable, please proceed in the following order. Each step should leave a short record of what changed and which results were affected.

1. **Correct the mathematical conventions.** Establish the actual mirror action, distinguish reversal from mirroring, verify the Khovanov coordinate and width conventions, and rerun the affected mirror summaries.

2. **Repair the statistical justification.** State the null hypotheses and exchangeability assumptions, review the residual test with statistical advice, and report effect sizes and movable counts across the amplitude grids. Decide which comparisons are descriptive and which have a justified inferential test.
3. **Complete the methods and comparisons.** Document percentile aggregation and the residual projection, assess the projection's influence, clarify the history of data reuse, and carry out the single-view and two-view comparisons. If a fresh validation sample is available, reserve it before inspecting its outcomes.
4. **Add the mathematical interpretation.** Include the distance–angle identity and the Turaev-genus consequence, verify the individual knots behind the counts, and find one useful example. Then investigate the symmetric score and the cancellation quantity with the limited purposes described above.
5. **Rewrite and reconcile the documents.** Revise the title, abstract, introduction, figures, and prose. Finally, make the manuscript, supplement, code, and saved numerical results agree, and complete the release and acknowledgements information before submission.

For the next version, please send the revised manuscript and supplement together with a concise response to these points and the exact code version that produced the tables. Where a proposed calculation changes the conclusion, tell us plainly. Where an addition proves uninformative, explain that as well. We can make the final decision about the paper's emphasis from those results.

My present assessment is that the computational work is substantial, the question is worthwhile, and the attribution to previous mathematics has improved. The new knot-theoretic content is still modest, and the statistical qualifications need to be resolved before the strongest claims are ready. I would be happy to see this become a clear, reliable paper for *Machine Learning: Science and Technology*, with a precise account of what the method teaches us. The additions above could make its mathematical purpose much easier to see.

Thank you again for the care you have put into the project. I hope these comments help you find the next steps. There is pleasure to be had in making an experiment precise enough that a mathematical question emerges from it. I would like us to bring this paper to that point.

With warm wishes,

Ernesto

References and code mentioned in the letter

- [1] D. Bar-Natan and R. van der Veen, *A Fast, Strong, Topologically Meaningful, and Fun Knot Invariant*. See Conjectures 20 and 21 in §6. [arXiv:2509.18456v4](https://arxiv.org/abs/2509.18456v4).
- [2] R. D. Shah and J. Peters, *The Hardness of Conditional Independence Testing and the Generalised Covariance Measure*, *Annals of Statistics* **48** (2020), 1514–1538. [arXiv:1804.07203](https://arxiv.org/abs/1804.07203).
- [3] T. Abe and K. Kishimoto, *The dealternating number and the alternation number of a closed 3-braid*. See Corollary 5.4 and Lemma 5.7, with the latter attributed there to Manturov and Champanerkar–Kofman–Stoltzfus. [arXiv:0808.0573](https://arxiv.org/abs/0808.0573).
- [4] Public code inspected: [EdisonVazquezG/knot_rare](https://github.com/EdisonVazquezG/knot_rare), commit [4d9b4fc](https://github.com/EdisonVazquezG/knot_rare/commit/4d9b4fc) (20 August 2026), especially stages 18B, 23, and 23B. Please reconcile this public version with the code used for the final manuscript.